

# UI & UX Design

SWE 4833

## Interaction Design

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# Interaction Design

- Interaction design is the design of the interaction between users and products.
- The interaction between a user and a product involves elements like aesthetics, motion, sound, space, and many more.

# Why Interaction Design is Important?

- Better User Experience
- Improves Usability
- Increases Engagement
- Builds Brand Loyalty
- Supports Accessibility

## Elements of Interaction Design

- **Images:** Pictures help users understand concepts that might be confusing with just text. They make the site more visually appealing.
- **Videos:** Videos guide users through content and can teach new things in an easy-to-follow way.
- **Buttons:** Buttons that users can click on make the website feel more dynamic and responsive, enhancing the user experience.
- **Animations:** Animations add movement and can make navigation more enjoyable by showing how things change or interact on the site.
- **Inputs:** Allowing users to enter information, like text or images, through forms makes the site more interactive and personal.

## 5 dimensions of interaction design

- 1D: Words
- 2D: Visual representations
- 3D: Physical objects or space
- 4D: Time
- 5D: Behaviour

## 5 dimensions of interaction design

- **1D: Words**
  - Words used in interactions, like button labels, should be meaningful and simple to understand.
  - Placing the right word at the right place makes the user's interaction smoother and easier.
  - They should communicate information to users, but not too much information to overwhelm the user.
  - The text should exactly be conveying its role or purpose.

## 5 dimensions of interaction design

- **2D: Visual representations**
  - This concerns graphical elements like images, typography and icons that users interact with.
  - Sometimes, visual representation can be powerful as texts.
  - For example, certain iconography like search icons are very much familiar to the users and without even looking at the texts, they would get to know its purpose.

## 5 dimensions of interaction design

- **3D: Physical objects or space**
  - The medium through which the user interacts with the product.
  - It could be a Mobile/tablet screen, computer mouse or keyboard, joystick, etc.
  - Understand the medium and design things that are easily doable with them.
  - For example, when a user is working on a task using the mouse, we can show a hover action to give a clickable feel, but when it comes to the touch screen, there is no chance for a hover action.

## 5 dimensions of interaction design

- **4D: Time**

- It mostly refers to media that changes with time (animation, videos, sounds).
- Motion and sounds play a crucial role in giving visual and audio feedback to users' interactions.
- Also consider the amount of time a user spends interacting with the product, like progress bar animation where user can see the progress of a particular process/operation.

## 5 dimensions of interaction design

- **5D: Behaviour**
  - The actual behavior of our application, it's how the previous dimensions define the interactions of a product.
  - Showing a success message with a summary when we complete a task, Swipe actions, etc
  - Behavior is the emotions and reactions that the user has when interacting with the system.

# Types of Interaction

- **Instructing**
  - The user tells the system what to do.
  - Clicking a button, Selecting a menu option, Entering a command
- **Conversing**
  - The user interacts with the system through dialogue.
  - Chatbots, Voice assistants, AI assistants

# Types of Interaction

- **Manipulating and Navigating**
  - The user directly interacts with objects or moves through a digital environment.
  - Dragging and dropping files, Zooming into a map, Swiping through images
- **Exploring and Browsing**
  - The user explores information or a digital environment.
  - Browsing an e-commerce website, Exploring Google Maps, Navigating through a virtual environment